Chapter 11 • Loci and Construction

Solved Exercise 11.2

1. A and B are 8.2 cm apart. Construct the locus 5 cm from A.

Mark A and B with AB=8.2 cm. Set the compass to 5 cm and draw the complete circle centred at A.

Every point on this circle, and only those points, is 5 cm from A.

Circle with centre A and radius 5 cm.

2. Construct the locus 2.2 cm from segment CD=5.7 cm.

Draw CD=5.7 cm. Construct parallel lines on both sides at perpendicular distance 2.2 cm.

For the complete distance-from- segment locus, join their ends with semicircles of radius 2.2 cm centred at C and D.

3. Construct ∠ABC=105° and the locus equidistant from BA and BC.

Draw BA and BC with included angle 105°. Bisect the angle using compass arcs.

For points inside the angle, the locus is the internal bisector. For the two infinite lines, the external angle bisector also satisfies equal perpendicular distances.

4. E and F are 5.4 cm apart. Construct the locus equidistant from E and F.

Draw EF=5.4 cm. With radius greater than 2.7 cm, draw equal arcs centred at E and F on both sides.

Join the two arc intersections; the line passes through the midpoint at 2.7 cm and is perpendicular to EF.

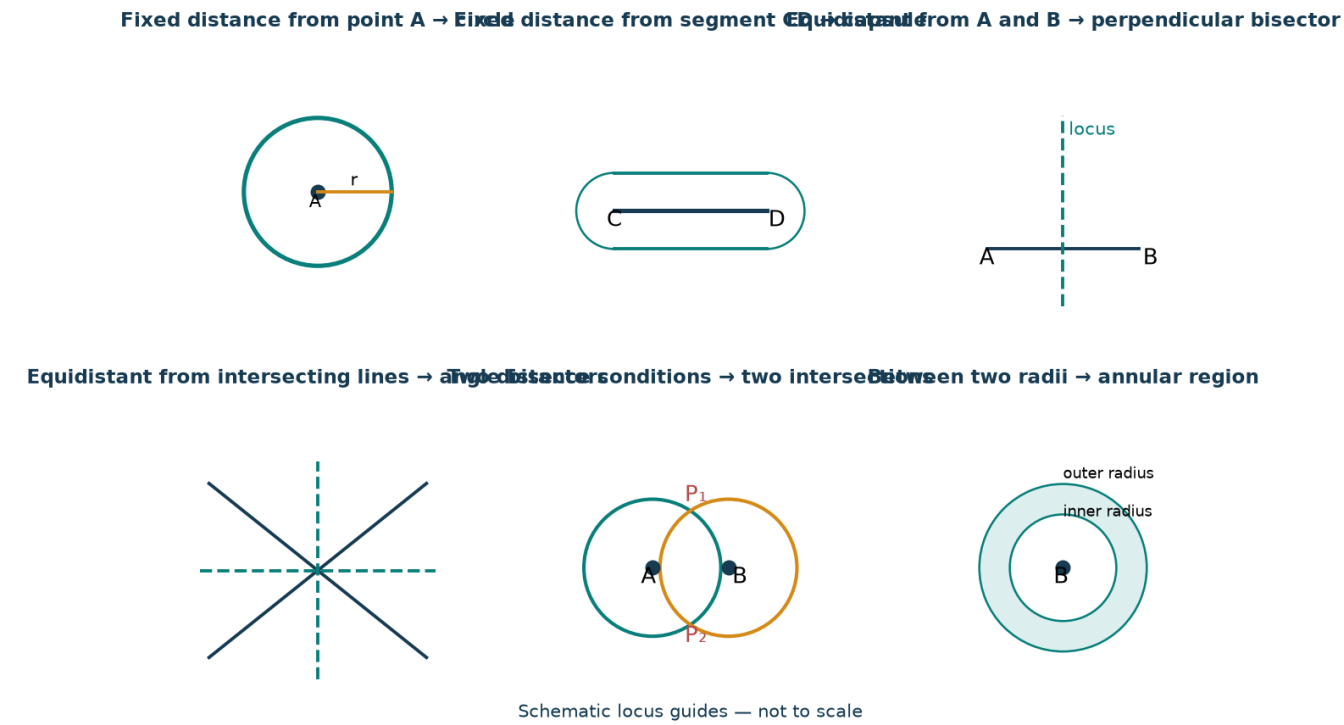
Perpendicular bisector of EF.

5. Cities A and B are 8 km apart; Kashif is 6.8 km from A and 7.3 km from B. Mark possible locations.

Choose a scale, for example 1 cm=1 km. Draw AB=8 cm, circle(A,6.8 cm) and circle(B,7.3 cm). Mark both intersections with crosses.

Along AB, the intersection foot is $x=\frac{6.8^2+8^2-7.3^2}{2\times 8}=3.559$ cm from A; perpendicular offset=$\sqrt{(6.8^2-x^2)}\approx 5.794$ cm.

The intersections exist because $7.3-6.8<8<7.3+6.8$.



Schematic locus guides — not to scale

6. Construct CD=7.6 cm, ∠D=45°, DE=5.9 cm; intersect the loci equidistant from C,D and from CD,CE.

Draw CD=7.6 cm, a 45° ray at D, mark DE=5.9 cm and join CE.

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(a) Construct the perpendicular bisector of CD
(b) Construct the internal angle bisector of ∠DCE.
Their intersection is X; label any second intersection with the external angle bisector separately only if full lines are requested.

7. Construct LM=7 cm, ∠L=70°, ∠M=45°. Find an interior point equidistant from L,M and 3 cm from L.

The triangle itself is constructible: ∠N=65°. The equal-distance locus from L and M is the perpendicular bisector of LM.
Every point P on that bisector satisfies PL=PM≥$\frac{LM}{2}$=3.5 cm. It cannot satisfy PL=3 cm.
No such point exists ; the distance condition contradicts LM=7 cm.

8. Construct right ΔRST with RS=6.8 cm, ∠S=90°, ST=7.5 cm. Find an interior point equidistant from RS,RT and 4.5 cm from R.

Draw perpendicular RS and ST with the stated lengths and join RT.
Bisect ∠SRT; this is the equal-perpendicular-distance locus from RS and RT.
Draw circle(R,4.5 cm). Its intersection with the internal bisector inside the angle is the required point P.
P is on the internal bisector of ∠SRT with RP=4.5 cm.

9. Construct rectangle UVWX with UV=7.2 cm, VW=5.6 cm. Draw loci 2 cm from UV and 3.5 cm from W.

Draw the rectangle. The distance-from-line-UV locus is the pair of parallel lines 2 cm above and below UV; use capsule ends if UV is treated strictly as a segment.
Draw circle(W,3.5 cm). The parallel line inside the rectangle is 5.6−2=3.6 cm from W, already greater than the circle radius; the outside parallel is even farther.

10. Draw the navigation locus equidistant from cell towers A and B.

Construct the perpendicular bisector of AB. Every point on it has equal distances to the two towers.

11. Draw the quarantine locus/region 10 km from an infection source O.

Using a suitable scale, draw circle(O,10 km) for points exactly 10 km away.
For “within 10 km”, shade the disk enclosed by the circle; the boundary is included if “within” means ≤10 km.
Boundary=circle radius 10 km; quarantine area=disk radius 10 km.

12. Treasure is 24 km from A and equidistant from B,C; scale 1 cm=10 km.

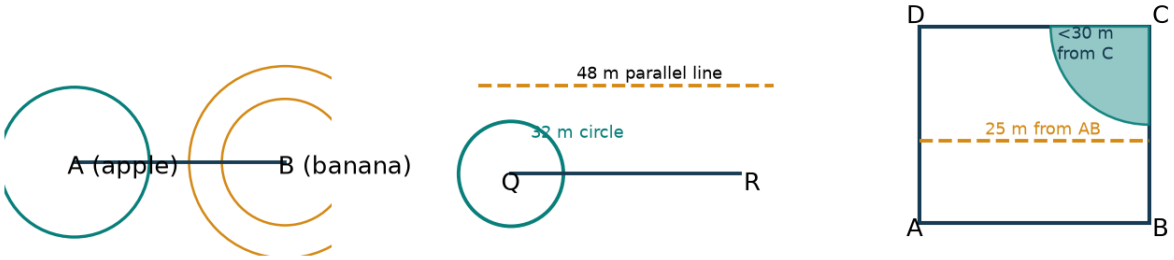
Draw the given relative positions of A,B,C. Construct circle(A,2.4 cm) and the perpendicular bisector of BC.
Their intersection(s) are the possible treasure locations. A line and circle can give 0, 1 or 2 points; exact locations cannot be selected unless the A,B,C positions/distances are supplied.
Intersect circle(A,2.4 cm) with the perpendicular bisector of BC; positional data are otherwise insufficient.

13. Apple tree A and banana tree B are 90 m apart. Mango M is 64 m from A and between 54 m and 82 m from B; scale 1 cm=10 m.

Draw AB=9 cm, circle(A,6.4 cm), circle(B,5.4 cm) and circle(B,8.2 cm).
The required positions are the portions of circle(A,6.4 cm) lying between the two B-centred circles, including endpoints if “between” is inclusive.
Each boundary circle meets the A-circle in two points, so the solution is two circular arcs , not just finitely many points.
Two arcs of circle(A,64 m) inside the annulus 54 m≤MB≤82 m.

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Mango tree: two arcs of the 64 m rule **Review 7: separate loci; no simultaneous point** **Review 8: required field region**



Schematic applications — not to scale; strict boundaries are dashed

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